

RADWAY AGCM, AB, Canada

WMO: 715450

Lat: 53.9950N

Lon: 112.8791W

Elev: 2051

StdP: 13.64

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-27.7	-21.4	-34.3	0.8	-27.6	-27.7	1.3	-21.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	82.6	63.2	79.6	61.8	76.7	60.8	66.9	77.6	64.6	75.2	62.7	72.8	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	92.2	72.8	60.6	85.2	69.6	58.7	79.4	67.4	32.6	78.0	30.8	75.3	29.3	72.9	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-36.6	88.5	6.9	1.6	-41.5	89.6	-45.5	90.6	-49.4	91.4	-54.4	92.6		
(5)			-36.7	70.8	6.8	3.4	-41.6	73.3	-45.6	75.3	-49.5	77.2	-54.4	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	11.3	10.5	22.1	37.8	51.5	58.4	62.1	60.0	51.6	38.7	22.6	9.4		
	DBStd	22.32	15.68	14.93	14.42	9.46	8.21	5.35	4.45	5.69	7.98	8.42	12.36	14.28		
	HDD50	6149	1200	1106	865	376	82	6	0	2	75	359	821	1257		
	HDD65	10461	1665	1526	1330	815	421	206	114	171	406	814	1271	1722		
	CDD50	1214	0	0	0	10	129	257	374	313	121	10	0	0		
	CDD65	51	0	0	0	0	3	7	22	17	2	0	0	0		
	CDH74	1270	0	0	0	5	153	167	389	374	179	3	0	0		
	CDH80	232	0	0	0	0	29	21	64	75	43	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	17.0	0.9	0.5	0.7	1.3	1.9	3.0	3.3	2.1	1.4	0.8	0.9	0.7		
(15) Precipitation	PrecMax	23.9	3.5	1.0	1.9	2.5	4.8	5.7	6.3	3.9	3.4	2.2	3.5	1.6		
	PrecMin	9.8	0.1	0.0	0.2	0.0	0.3	1.0	0.9	0.4	0.2	0.0	0.1	0.2		
	PrecStd	2.8	0.7	0.3	0.4	0.7	0.9	1.3	1.2	0.9	0.8	0.5	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	44.0	54.7	73.3	84.1	82.7	85.6	86.4	85.8	71.9	53.0	42.2		
		MCWB	37.3	37.3	43.9	51.0	57.4	62.5	67.3	63.6	61.8	53.4	43.6	35.6		
	2%	DB	40.4	39.1	47.5	65.4	78.3	78.8	81.5	81.6	79.1	62.6	45.8	37.7		
		MCWB	35.3	34.1	39.0	47.0	54.3	60.9	65.8	64.2	59.0	48.8	39.3	32.7		
	5%	DB	37.6	35.6	43.1	59.5	73.7	75.0	78.4	78.0	73.9	57.2	41.6	33.9		
		MCWB	33.6	32.1	36.7	44.6	52.8	58.3	64.3	62.7	56.8	45.4	36.2	30.2		
	10%	DB	34.1	32.3	39.4	54.0	69.4	71.7	75.3	74.9	68.0	52.6	37.7	29.5		
		MCWB	31.0	29.5	34.3	41.7	51.0	57.3	62.2	61.5	54.3	43.6	34.3	26.8		
	0.4%	WB	37.8	38.0	44.4	51.9	58.4	65.9	71.0	69.1	63.2	54.4	44.5	36.5		
		MCDB	42.9	43.7	52.9	72.5	76.6	77.0	80.1	78.9	79.2	70.1	51.1	41.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	34.7	40.1	47.8	56.3	63.0	67.8	66.5	60.3	50.2	40.0	33.0		
		MCDB	39.8	38.1	47.0	63.9	72.6	73.2	78.4	77.6	76.5	59.5	44.9	37.3		
	5%	WB	33.7	32.3	37.1	45.0	54.5	60.8	65.5	64.0	57.7	47.0	37.1	30.1		
		MCDB	37.4	35.3	43.0	57.8	70.5	72.3	75.6	74.5	72.0	55.2	40.9	33.6		
	10%	WB	30.9	29.2	34.5	42.5	52.5	59.0	63.5	62.2	55.4	44.5	34.2	26.9		
		MCDB	34.1	32.3	38.8	52.4	66.6	69.1	73.0	72.8	66.4	51.6	37.4	29.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	20.2	19.0	20.9	25.4	21.8	22.7	24.6	25.3	20.2	16.1	16.9		
		MCDBR	19.2	22.2	21.7	31.3	33.2	27.9	28.7	30.8	37.1	28.4	19.8	19.8		
		MCWBR	15.4	17.9	14.5	15.8	14.2	13.0	15.1	15.0	17.8	16.2	13.8	16.3		
		MCDBR	18.7	20.7	20.9	28.6	30.1	23.7	26.1	27.0	34.2	27.2	18.4	20.1		
	5% WB	MCWBR	15.2	17.1	14.2	15.2	13.2	12.1	14.7	14.3	16.9	16.1	13.5	16.8		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.267	0.310	0.329	0.342	0.356	0.355	0.364	0.308	0.281	0.252	0.241		
	taud		2.291	2.327	2.292	2.374	2.391	2.367	2.414	2.342	2.501	2.548	2.448	2.280		
	Ebn at Noon		229	267	276	286	287	283	281	271	277	263	235	210		
	Edh at Noon		18	24	32	34	35	36	34	34	26	20	15	15		
(44) All-Sky Solar Radiation	RadAvg		288	615	1100	1479	1808	1896	1928	1584	1092	620	293	211		
	RadStd		33	46	83	130	142	125	99	111	112	78	32	19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page